Constructing bottom-up government spending projections

A walk-through of allocations, demographic demand, cost assumptions and factor contributions for Australian consolidated general government.

Status: research scenario model. The expansion creates 24 separately projected blocks, including separate higher education and VET, with explicit proxies and residuals. Health/education now use distinct cost rules and a common aggregate multiplier. It does not establish causal productivity elasticities or reproduce official programme forecasts. The report uses default assumptions; the Shiny recalculates the same decomposition for selected scenarios.

1. What has changed, and what remains an assumption

The initial prototype used one demand index per broad category. The first expansion introduced separate health age profiles and named spending blocks. This update adds distinct health/education input-cost rules, separates higher education and VET, and replaces their aggregate productivity-exponent overrides with common annual cost multipliers. Existing NDIS and pension paths, the fiscal engine and all 2030 category totals are retained.

This is a hybrid bottom-up model. Published allocations and age patterns anchor the estimates, while coverage, cost response, intensity and some demand profiles remain assumptions. A detailed name is not evidence of a detailed behavioural model. Residential/home care still share an age profile; other social protection remains a working-age residual. These are priorities for subsequent refinement.

Item	2025 ($bn)
Health named government-funding blocks	173.90
Health residual and GFS reconciliation	33.84
Other social protection, still unallocated	163.39

2. Accounting perimeter and construction sequence

The perimeter is consolidated Australian general government. ABS national totals are the accounting control. Commonwealth grants to states, GST transfers and contributions to NDIS funding are internal transfers, not extra national spending. AIHW government-funder amounts are used only to allocate named health blocks; state own-source funding is combined with Commonwealth funding without adding grants again. Private household and insurance funding are excluded from allocations.

Stage	Construction
1	2025 ABS functional primary expense controls, removing conventional interest once
2	Allocate named subfunctions using ABS education, AIHW government health funding and GEN care amounts; retain explicit residuals
3	Project subfunction allocation weights to 2030 using demographic exposures and the existing bridge assumptions

Stage	Construction
4	Reconcile within each category to its unchanged 2030 model anchor; all categories still match PBO aggregate primary expenses
5	Project each block from its own 2030 allocation using consistent multiplicative factors
6	Sum blocks to categories; add modelled interest and net capital investment in the financing engine

ABS primary expenses include depreciation and capital transfers where classified as expenses. Gross capital formation is not added again. The model adds net capital investment from the PBO anchor separately; expenses plus net investment is the fiscal-balance spending measure. Source concepts differ: AIHW recurrent funding, programme cash outlays and GFS accrued expenses do not perfectly align. Reconciliation is an allocation device, not proof of conceptual equivalence.

3. The source-to-anchor bridge

The aggregate 2030 anchor comes from the PBO 2026-27 National Fiscal Outlook. Individual category anchors are inherited e61 allocations, not published PBO category forecasts. The prior category bridge used 2025 GFS/programme outlays, exposure growth, 2.5% price growth, 1.2% productivity growth, programme pass-through, 0.3% extra service prices where applicable and programme intensity; one common factor matched PBO primary expenses. This bridge is retained to isolate the effects of the subfunction expansion.

Within a category, the new 2030 allocation is proportional to its 2025 subfunction amount times its own 2025-30 demographic exposure growth and the bridge cost factors. A category-specific reconciliation factor then makes the sum equal the unchanged parent anchor. Changing a scenario does not recalibrate any 2030 amount.

```
raw block[2030] = block[2025] x demand[2030]/demand[2025] x bridge price/cost factors
block anchor[2030] = raw block[2030] / sum(parent raw blocks[2030]) x parent anchor[2030]
```

Category	Subfunction	2025 source/reconciled ($bn)	2030 allocation ($bn)	Share of parent in 2030 (%)
Health	Hospital services	103.20	128.89	49.98
Health	Primary and community care, excluding PBS medicines	36.80	45.79	17.75
Health	Referred medical services	18.10	22.52	8.73
Health	PBS medicines outside hospitals	15.80	19.76	7.66
Health	Other health and GFS reconciliation	33.84	40.92	15.87
Education	School education	80.66	88.59	53.10
Education	Higher education	43.71	50.36	30.18
Education	Vocational education and training	7.73	8.93	5.35
Education	Other education	16.34	18.96	11.37
Aged care	Residential aged care	24.20	33.64	60.80
Aged care	Home care and support	12.90	17.93	32.41

Category	Subfunction	2025 source/reconciled ($bn)	2030 allocation ($bn)	Share of parent in 2030 (%)
Aged care	Flexible aged care	1.20	1.67	3.02
Aged care	Other aged care	1.50	2.08	3.77
Age Pension	Age Pension	62.20	78.46	100.00
NDIS	NDIS participant plans	46.35	56.34	100.00
Other social protection	Other social protection (unallocated programmes)	163.39	190.34	100.00
Other public services	General public services excluding conventional interest	64.03	75.17	24.16
Other public services	Public order and safety	52.77	61.95	19.92
Other public services	Economic affairs	46.37	54.43	17.50
Other public services	Environmental Protection	20.78	24.39	7.84
Other public services	Housing and community amenities	13.90	16.32	5.25
Other public services	Recreation, culture and religion	19.97	23.44	7.54
Other public services	Transport	47.16	55.36	17.80
Defence	Defence	50.17	55.65	100.00

4. A common decomposition

```
block spending[t] = anchor[2030] x population x age/cohort demand x coverage
                    x general prices x real unit cost/payment
                    x relative service prices x service intensity x policy/financing rule
                    x aggregate cost multiplier
spending share[t] = 100 x block spending[t] / nominal GDP[t]
```

Factors equal one in 2030. For ordinary services, population times age/cohort demand equals the age-weighted service exposure. Fixed age weights assume unchanged age-specific relative need, utilisation and service mix. They do not model healthy ageing, migration composition or disability severity. Population controls uniformly scale the sourced age/sex shares; employment uses the same total-population path.

Factor	Definition
Population size	Total population relative to 2030; one for policy-target and interest blocks.
Age / cohort demand	Age-weighted exposure divided by total population. NDIS includes cohort retention, not a new disability-incidence forecast.

Factor	Definition
Coverage / eligibility	Pension recipient coverage relative to 2030. One for other blocks: constant eligibility/uptake is a maintained assumption.
General prices	Domestic price index inferred from nominal GDP, real output and ToT; one for target/interest rules. In real-dollar analysis this factor includes deflation.
Real unit cost / payment	Health/education: separate input baskets or funding proxies divided by service productivity. Other services: productivity exponents. Pension: mean real payment. Investment: exponent 0.5.
Relative service prices	Additional service price factor, 0.3% a year by default; excluded from medicine CPI and VET funding proxies.
Service intensity	Additional programme intensity growth: Health 0.5%, aged care 0.3%, NDIS 0.5%, others zero. Guesses, not fitted residuals.
Policy / financing rule	Defence spending follows its GDP-share target. Interest follows gross-debt financing. This factor captures the whole rule, not an identified policy shock.
Aggregate cost scaling	Health/education common annual cost adjustment compounded after 2030; zero by default, preserving subfunction differences.
GDP denominator	2030 GDP divided by current GDP. Used only for GDP-share contributions; the nominal income denominator must be accounted for.

Domestic prices are inferred from GDP = real output x domestic prices x the ToT income factor. Higher NGDP/person at unchanged real assumptions scales nominal service spending and GDP together. Hospital, school and university input weights are observed shares applied as scope proxies; VET uses a published funding mix as a proxy. Remaining weights, future coverage, service productivity and intensity are disclosed judgments. None is a causal productivity elasticity.

Health/education cost rules and evidence

Subfunction	Rule	Wage weight (%)	Status
Higher education	input_mix	55.47	Observed input share; scope proxy
Other education	input_mix	80.00	Retained judgment
School education	input_mix	75.99	Observed input share; scope proxy
Vocational education and training	funding_proxy	69.75	Published funding rule; scope proxy
Hospital services	input_mix	67.25	Observed input share; scope proxy
PBS medicines outside hospitals	cpi_proxy	0.00	Explicit judgment
Other health and GFS reconciliation	input_mix	70.00	Retained judgment
Primary and community care, excluding PBS medicines	input_mix	70.00	Retained judgment
Referred medical services	input_mix	70.00	Retained judgment

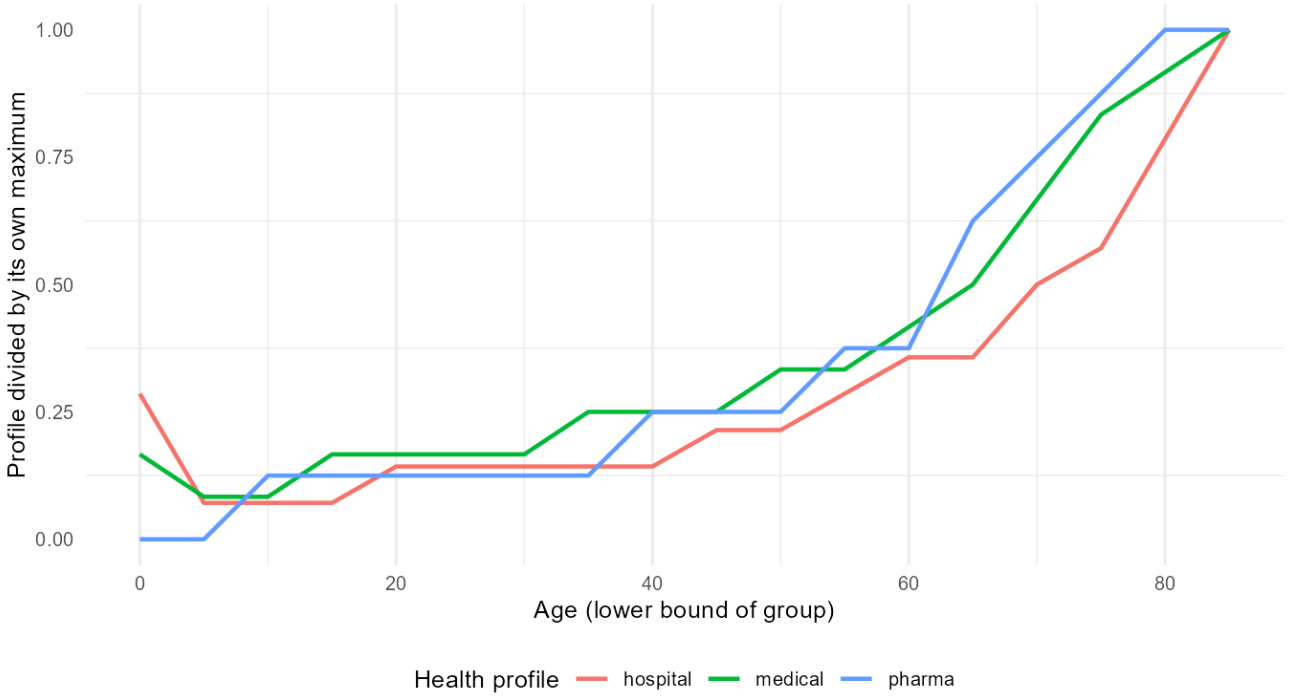
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Input basket: real unit cost = [(1-L) + L x real hourly wage index] / service-productivity index
VET: real unit cost = (1 + 0.6975 x annual productivity growth)^years / service-productivity index
Aggregate multiplier = (1 + selected annual cost adjustment)^years; default adjustment = 0
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Hospital labour share is 67.253% from IHACPA public admitted-acute national costs; school labour share is 75.993% from ACARA recurrent expenditure excluding notional user cost of capital; university employee share is 55.473% from Finance 2024. VET weight 69.75% derives from the National Skills Agreement flexible-funding formula; its productivity discount is assumed zero. These applications extend beyond exact source coverage. The active nine-row file data/health_education_parameters.csv records each rule, source locator and limitation. The README provides full arithmetic, linked references and reproduction commands.

5. Health: separate profiles and a reconciliation block

Hospital allocation uses Commonwealth public/private hospital funding plus state hospital funding. Primary/community care combines Commonwealth and state primary care, subtracting PBS medicines once because those medicines are separated. Referred medical services and outside-hospital PBS medicines are separate blocks. Other health and GFS reconciliation is the ABS Health total less these named blocks; research, administration and conceptual differences are not silently assigned a hospital profile.

Hospital services use the IGR C12.10 public-hospital age profile; applying it to private hospitals is a proxy. Primary and referred care use the medical-benefits profile, including a proxy application to community/public-health services. Medicines use the pharmaceutical-benefits profile. Other health follows total population. The three published shapes are used as relative weights, not as independently validated national unit-cost levels.



Distinct published age profiles, each normalised to its own maximum for visual comparison. Source: 2026 IGR C12.10. The 85+ source group is carried to age 100+.

Worked example: hospital services

Step	Value
2025 hospital government funding	$103.20bn
2030 calibrated hospital allocation	$128.89bn
Population factor	1.5364
Age/cohort factor	1.1623

Step	Value
General price factor	2.0269
Real unit-cost factor	1.3607
Relative service-price factor	1.1139
Intensity factor	1.1967
Aggregate cost multiplier	1.0000
2066 hospital spending	$846.18bn

Multiply the 2030 hospital allocation by the seven displayed growth factors to recover the 2066 amount. Coverage is one. The admitted-acute labour share is applied as a proxy for all hospital funding. Primary, referred and residual health retain a 70% judgmental wage weight; medicines use CPI-only unit costs with no service-price premium. Health intensity remains 0.5% throughout, unestimated. The aggregate cost slider multiplies every unit-cost path equally and preserves their differences.

6. Education: school, tertiary and other demand

ABS Education Table_1 supplies consolidated operating expenses for school, tertiary and other education. These sum to the GFS Education control; school grants are not added separately. School demand is population aged 5-17, higher education ages 18-24 and VET ages 15-64. These are proxies with fixed enrolment/coverage, not fitted participation models. Other education, including preschool and residual services, follows total population. Mature-age and international enrolment changes are not separately projected.

Higher education and VET now split the ABS tertiary total using 2024 university expenses ($43.068392bn) and VET recurrent expenditure excluding user cost of capital ($7.6172bn) as allocation weights. Concepts differ: this is a proxy, not a directly observed GFS split. Higher-education demand remains ages 18-24; VET uses ages 15-64. Schools and universities use separate observed input baskets as scope proxies; VET uses the funding proxy. Other education retains an 80% judgmental wage weight. Coverage and service productivity are constant; education intensity is zero. The inherited category bridge remains unchanged.

7. Care, transfers and other services

GEN 2024-25 amounts split aged care into residential, home/support, flexible and other care. Each currently uses the same IGR aged-care age profile and parent cost rule. The split makes allocations and coverage transparent but does not yet create separate care-uptake forecasts. The existing 0.3% aged-care intensity and 80% pass-through remain in place.

NDIS remains one participant-plan block, with the existing below-65 participation rates and retention of older cohorts. The user-selected 85% pass-through and 0.5% intensity are unchanged. Population versus age/cohort contributions separate overall scale from the participant/population ratio. The latter includes cohort retention; it should not be interpreted as pure demographic ageing or new eligibility-policy change.

Age Pension retains the representative means-test and private-wealth model. Its factors are eligible age demand, recipient coverage, domestic prices and mean real payment per recipient. Changes in means tests, retirement assets and recipient composition can affect both coverage and mean payment. These are accounting channels, not separate causal estimates of indexation or super returns.

Other social protection remains the GFS residual after removing pension, care and NDIS. Its demand is ages 15-66 and its existing 65% pass-through is retained; constant-relative-standards policy overrides that to 100%. It is not a simulation of the statutory indexation of individual benefits. The residual is material and should be split into disability, unemployment, family/childcare and other programmes before making stronger policy claims.

Other public services now sum seven named GFS functions, including general public services net of conventional interest, public order/safety, economic affairs, environment, housing/community amenities, recreation/culture/religion and transport. All retain total-population demand and the common parent cost rule. Separate names identify budget weights; they do not establish workload forecasts. Interest on defined-benefit superannuation remains inside the GFS residual, consistent with the prior model.

Defence is a target rule reaching 2.5% GDP by 2035. Its demographic and price factors are set to one and the policy factor reproduces the target path. Interest is represented by a financing factor reproducing the actual model bill; it combines gross debt and the nominal borrowing rate. Investment uses population, prices and a 0.5 productivity exponent. These explicit exceptions avoid mislabelling financing and targets as demographic service demand.

8. Contributions to the change: method and interpretation

A percentage change in a factor is not its dollar contribution because factors interact. We use exact Shapley accounting: average each factor's marginal contribution across all orders of switching factors from the start-year to end-year values. This allocates interactions symmetrically, handles zero-valued factors and sums exactly to each block's observed model change. Contributions are summed from blocks to categories and national totals. They are not regression coefficients, causal effects or confidence intervals.

For example, a $100 starting amount with population increasing 10% and prices 20% ends at $132. The method assigns $11 to population and $21 to prices, sharing the $2 interaction. Factors which do not change receive zero. Contributions to a GDP-share change include the GDP denominator as a separate factor. This is why population and prices can have positive numerator contributions alongside a negative denominator contribution.

Nominal-dollar contributions describe the actual projected dollar increase. Real-dollar contributions divide by the domestic price index, in 2030 prices; the general-price factor cancels for ordinary price-indexed services. This is a model deflator, not a separately measured sector output-price index. GDP-share contributions describe percentage-point changes in fiscal burden. Always choose the measure that answers the question being considered.

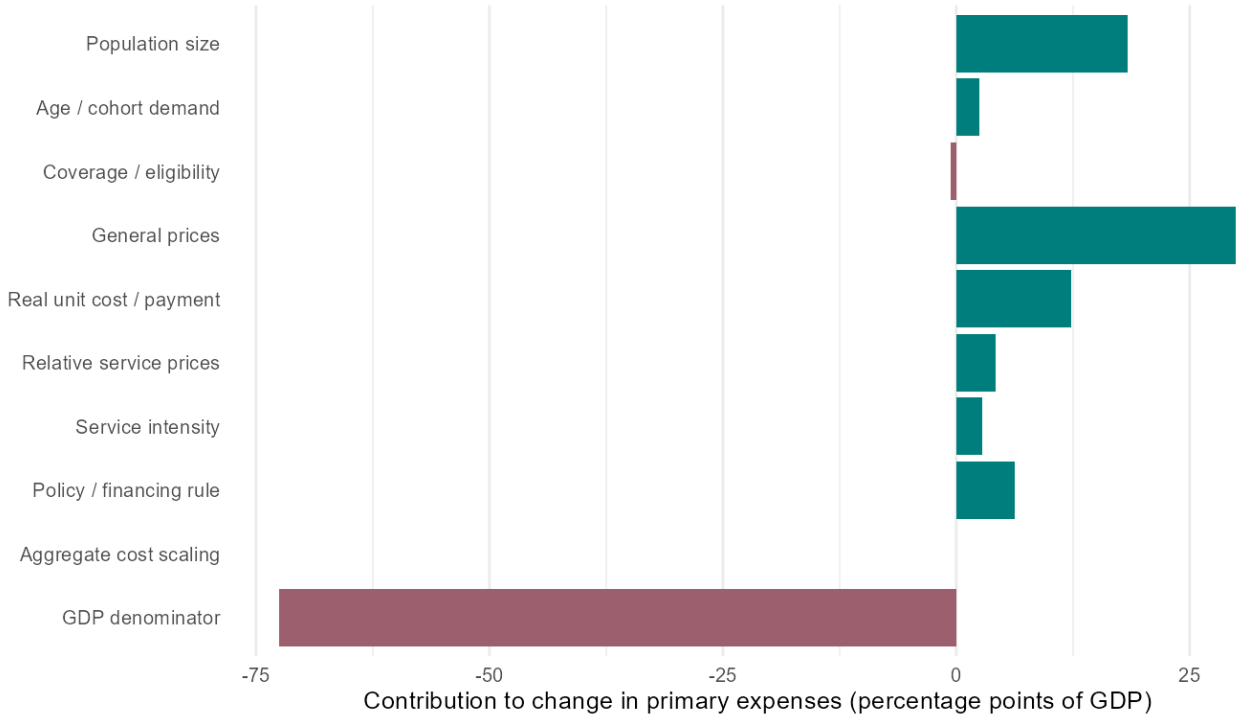
Primary spending contributions

Factor	Nominal change ($bn)	Real change ($2030 bn)	GDP-share change (pp)
Population size	1,211.564	779.032	18.298
Age / cohort demand	166.562	105.310	2.427
Coverage / eligibility	-44.470	-28.183	-0.651
General prices	1,924.036	-120.161	29.928
Real unit cost / payment	829.362	528.966	12.315
Relative service prices	292.175	184.458	4.244
Service intensity	190.589	119.428	2.724
Policy / financing rule	363.040	271.074	6.259
Aggregate cost scaling	0.000	0.000	0.000

Factor	Nominal change ($bn)	Real change ($2030 bn)	GDP-share change (pp)
GDP denominator	0.000	0.000	-72.504

All spending, including interest and net capital investment

Factor	Nominal change ($bn)	Real change ($2030 bn)	GDP-share change (pp)
Population size	1,260.508	810.985	19.060
Age / cohort demand	166.562	105.310	2.427
Coverage / eligibility	-44.470	-28.183	-0.651
General prices	2,002.203	-359.748	31.183
Real unit cost / payment	854.403	545.125	12.696
Relative service prices	292.175	184.458	4.244
Service intensity	190.589	119.428	2.724
Policy / financing rule	1,154.422	861.982	19.903
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-85.853

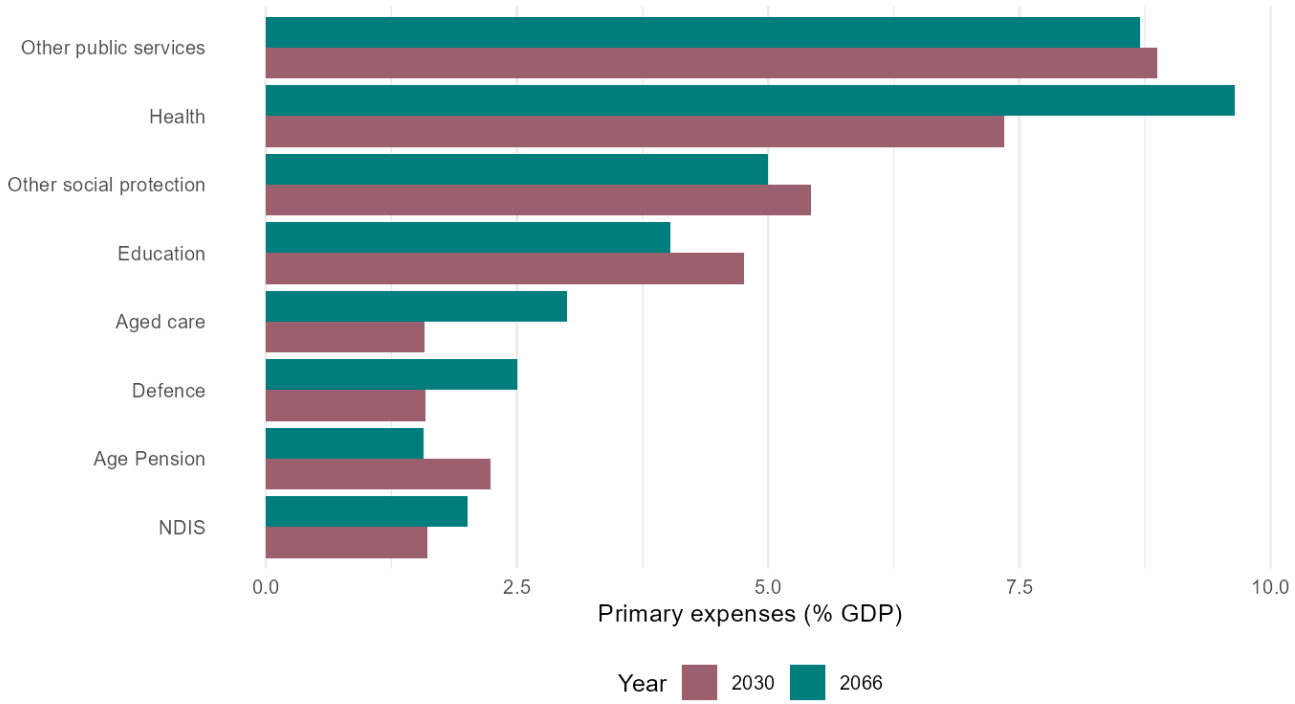


Exact factor accounting for the change in primary spending/GDP, 2030-2066. Positive numerator effects and the denominator effect should be read together.

9. Default projection and revision from the previous model

Assumption	Value
Projection window	Financial years ending 2030 to 2066
Population growth	1.2% a year
NGDP per person growth	3.2% a year
Productivity growth	1.2% a year
Average hours growth	0% a year
Relative service prices	0.3% a year
Tax thresholds	Wage indexed
Real super return	4% a year
Long-run nominal borrowing rate	4.4%

Measure	2030	2066
Nominal GDP ($bn)	3,507.41	16,747.68
Primary expenses ($bn)	1,171.90	6,104.76
Interest expenses ($bn)	77.20	868.58
Net capital investment ($bn)	53.20	205.35
Total spending including net investment ($bn)	1,302.30	7,178.69
Primary expenses (% GDP)	33.41	36.45
Total spending including net investment (% GDP)	37.13	42.86
Net debt (% GDP)	36.47	112.28



The category totals are sums of separately projected subfunctions; the starting category anchors are fixed.

Category	2030 ($bn)	2066 ($bn)	2030 (% GDP)	2066 (% GDP)	Change (pp)
Age Pension	78.46	263.60	2.24	1.57	-0.66
Aged care	55.32	502.15	1.58	3.00	1.42
Defence	55.65	418.69	1.59	2.50	0.91
Education	166.84	674.03	4.76	4.02	-0.73
Health	257.89	1,615.43	7.35	9.65	2.29
NDIS	56.34	336.31	1.61	2.01	0.40
Other public services	311.05	1,457.32	8.87	8.70	-0.17
Other social protection	190.34	837.23	5.43	5.00	-0.43

The next table compares this update with the frozen 23-block model immediately before the health/education cost changes, using identical macro and tax-policy defaults and neutral cost controls in each version. Distinct input-cost assumptions and the separate higher/VET allocation and demand revise spending. The other six primary category paths are unchanged. Debt and interest change as a consequence. The revision does not establish forecast accuracy.

Category	Previous 2066 (% GDP)	Expanded 2066 (% GDP)	Revision (pp)
Age Pension	1.574	1.574	0.000
Aged care	2.998	2.998	0.000
Defence	2.500	2.500	0.000
Education	4.128	4.025	-0.104
Health	9.798	9.646	-0.153

Category	Previous 2066 (% GDP)	Expanded 2066 (% GDP)	Revision (pp)
NDIS	2.008	2.008	0.000
Other public services	8.702	8.702	0.000
Other social protection	4.999	4.999	0.000

10. Category-by-category factor contributions

Health

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	318.513	202.562	4.700
Age / cohort demand	97.313	60.979	1.391
Coverage / eligibility	0.000	0.000	0.000
General prices	503.768	0.000	7.654
Real unit cost / payment	221.385	139.795	3.217
Relative service prices	78.579	49.111	1.117
Service intensity	137.986	86.651	1.981
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-17.766

Education

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	158.761	103.391	2.462
Age / cohort demand	-62.899	-39.695	-0.913
Coverage / eligibility	0.000	0.000	0.000
General prices	253.319	0.000	4.047
Real unit cost / payment	118.897	76.964	1.822
Relative service prices	39.112	25.036	0.585
Service intensity	0.000	0.000	0.000
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-8.736

Age Pension

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	67.381	44.333	1.067
Age / cohort demand	48.793	31.889	0.763
Coverage / eligibility	-44.470	-28.183	-0.651
General prices	107.926	0.000	1.761
Real unit cost / payment	5.509	3.548	0.084
Relative service prices	0.000	0.000	0.000
Service intensity	0.000	0.000	0.000
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-3.685

Aged care

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	87.867	54.858	1.246
Age / cohort demand	103.067	64.597	1.474
Coverage / eligibility	0.000	0.000	0.000
General prices	138.037	0.000	2.014
Real unit cost / payment	71.305	44.337	1.002
Relative service prices	23.274	14.312	0.319
Service intensity	23.274	14.312	0.319
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-4.952

NDIS

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	67.759	43.206	1.006
Age / cohort demand	-0.295	-0.184	-0.004
Coverage / eligibility	0.000	0.000	0.000
General prices	107.274	0.000	1.639
Real unit cost / payment	58.112	36.933	0.856

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Relative service prices	17.790	11.160	0.255
Service intensity	29.329	18.465	0.424
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-3.774

Other social protection

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	189.560	122.867	2.911
Age / cohort demand	-19.416	-12.276	-0.283
Coverage / eligibility	0.000	0.000	0.000
General prices	301.929	0.000	4.776
Real unit cost / payment	125.396	80.629	1.894
Relative service prices	49.420	31.490	0.732
Service intensity	0.000	0.000	0.000
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-10.459

Other public services

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	321.723	207.815	4.906
Age / cohort demand	0.000	0.000	0.000
Coverage / eligibility	0.000	0.000	0.000
General prices	511.783	0.000	8.038
Real unit cost / payment	228.757	146.760	3.439
Relative service prices	84.000	53.349	1.236
Service intensity	0.000	0.000	0.000
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-17.786

Defence

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	0.000	0.000	0.000
Age / cohort demand	0.000	0.000	0.000
Coverage / eligibility	0.000	0.000	0.000
General prices	0.000	-120.161	0.000
Real unit cost / payment	0.000	0.000	0.000
Relative service prices	0.000	0.000	0.000
Service intensity	0.000	0.000	0.000
Policy / financing rule	363.040	271.074	6.259
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-5.346

Interest

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	0.000	0.000	0.000
Age / cohort demand	0.000	0.000	0.000
Coverage / eligibility	0.000	0.000	0.000
General prices	0.000	-239.587	0.000
Real unit cost / payment	0.000	0.000	0.000
Relative service prices	0.000	0.000	0.000
Service intensity	0.000	0.000	0.000
Policy / financing rule	791.381	590.907	13.644
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-10.659

Net capital investment

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Population size	48.944	31.953	0.763
Age / cohort demand	0.000	0.000	0.000
Coverage / eligibility	0.000	0.000	0.000
General prices	78.167	0.000	1.255
Real unit cost / payment	25.041	16.159	0.381

Factor	Nominal $bn	Real $2030 bn	GDP share pp
Relative service prices	0.000	0.000	0.000
Service intensity	0.000	0.000	0.000
Policy / financing rule	0.000	0.000	0.000
Aggregate cost scaling	0.000	0.000	0.000
GDP denominator	0.000	0.000	-2.689

11. Source and assumption audit by subfunction

Sourced means a published amount or shape. Calibrated means an allocation reconciled to a control total. Proxy means an assumed relationship to an available population/base. Judgment means an unestimated future response. Identity means an accounting consequence. The following entries disclose the role of each source and the additional assumptions; citing a source for an amount does not validate its future growth rule.

h_hospital: Hospital services

Source and extraction: https://www.aihw.gov.au/reports/health-welfare-expenditure/health-expenditure-australia-2024-25/contents/spending-trends-by-source/australian-government-spending ; Figures 11 and 16: Commonwealth public 35.4 + private 8.3 + state hospital 59.5 .

Allocation and demand: 2025 $103.20bn; reconciled $103.20bn; raw 2030 $139.20bn; final 2030 $128.89bn. Demand profile: hospital. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.925962.

Evidence status: Sourced allocation; model demand proxy . Rounded AIHW funding amounts; public-hospital age profile used for both public and private hospitals; no grants added separately .

Cost rule: input_mix ; wage weight 67.253 %; Observed input share; scope proxy . Appendix 5.2, national 2023-24 column: salary/wage line items plus oncosts / total line-item cost .

Cost source and limitation: https://www.ihacpa.gov.au/resources/national-hospital-cost-data-collection-public-sector-2023-24 ; Public admitted-acute cost mix applied to all hospital funding; includes VMOs and oncosts; does not identify future efficiency .

h_primary: Primary and community care, excluding PBS medicines

Source and extraction: https://www.aihw.gov.au/reports/health-welfare-expenditure/health-expenditure-australia-2024-25/contents/spending-trends-by-source/australian-government-spending ; Figures 11/16: Commonwealth primary 39.7 + state primary 12.9 - Commonwealth PBS within primary 15.8 .

Allocation and demand: 2025 $36.80bn; reconciled $36.80bn; raw 2030 $49.45bn; final 2030 $45.79bn. Demand profile: medical. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.925962.

Evidence status: Sourced allocation; model demand proxy . Medical-benefits age profile is a proxy for heterogeneous community/public-health services .

Cost rule: input_mix ; wage weight 70.000 %; Retained judgment . Previous 70% health exposure retained as an unestimated input-mix proxy; not the unpublished WCI-5 formula .

Cost source and limitation: Retained explicit scenario judgment ; Community/public health and Medicare funding need separate models; not measured labour share .

h_referred: Referred medical services

Source and extraction: https://www.aihw.gov.au/reports/health-welfare-expenditure/health-expenditure-australia-2024-25/contents/spending-trends-by-source/australian-government-spending ; Figure 11: referred medical services; state page notes no referred medical spending .

Allocation and demand: 2025 $18.10bn; reconciled $18.10bn; raw 2030 $24.32bn; final 2030 $22.52bn. Demand profile: medical. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.925962.

Evidence status: Sourced allocation; model demand proxy . IGR medical-benefits age profile used as a proxy .

Cost rule: input_mix ; wage weight 70.000 %; Retained judgment . Previous 70% health exposure retained; no claimed estimate of future MBS indexation .

Cost source and limitation: Retained explicit scenario judgment ; Government rebates differ from provider costs and include policy changes; not measured labour share .

h_medicines: PBS medicines outside hospitals

Source and extraction: https://www.aihw.gov.au/reports/health-welfare-expenditure/health-expenditure-australia-2024-25/contents/spending-trends-by-source/australian-government-spending ; Area of spending: PBS excluding drugs allocated to hospitals .

Allocation and demand: 2025 $15.80bn; reconciled $15.80bn; raw 2030 $21.34bn; final 2030 $19.76bn. Demand profile: pharma. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.925962.

Evidence status: Sourced allocation; model demand proxy . Does not separately model medicine innovation, patents, utilisation or price negotiations .

Cost rule: cpi_proxy ; wage weight 0.000 %; Explicit judgment . Government cost per prescription follows CPI; no extra real price trend fitted .

Cost source and limitation: Retained explicit scenario judgment ; Ignores medicine innovation, price disclosure, confidential rebates and future mix; 0.5% intensity retained, not estimated .

h_other: Other health and GFS reconciliation

Source and extraction: ABS GFS Health control less named AIHW funding blocks ; ABS Table 4; residual identity .

Allocation and demand: 2025 $33.84bn; reconciled $33.84bn; raw 2030 $44.19bn; final 2030 $40.92bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.925962.

Evidence status: Residual allocation; population proxy . Includes research, other services, classification/rounding and accounting differences; not an independently measured service group .

Cost rule: input_mix ; wage weight 70.000 %; Retained judgment . Previous 70% health exposure retained for heterogeneous residual .

Cost source and limitation: Retained explicit scenario judgment ; Includes reconciliation and research; not a single observed service .

e_school: School education

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25/55120DO095_202425.xlsx ; Table_1 row 9 2024-25 .

Allocation and demand: 2025 $80.66bn; reconciled $80.66bn; raw 2030 $97.82bn; final 2030 $88.59bn. Demand profile: school_age. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.905694.

Evidence status: Sourced allocation; model demand proxy . Ages 5-17 proxy; constant enrolment/coverage; includes government funding of all school sectors .

Cost rule: input_mix ; wage weight 75.993 %; Observed input share; scope proxy . ACARA Table 9.3: salaries including redundancies / recurrent spending excluding user cost of capital .

Cost source and limitation: https://dataandreporting.blob.core.windows.net/anrdataportal/ANR-Documents/ANR2024/nationalreportonschoolinginaustralia_2024_Chapter9.pdf ; Government-school cost structure applied to all government school funding; funding rules differ, oncost scope differs .

e_higher: Higher education

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25/55120DO095_202425.xlsx; Department of Education Finance 2024 Table 4; RoGS 2026 Table 5A.1 ; ABS tertiary control allocated using 2024 university total expenses versus VET recurrent expenses excluding user cost of capital .

Allocation and demand: 2025 $43.71bn; reconciled $43.71bn; raw 2030 $55.60bn; final 2030 $50.36bn. Demand profile: tertiary_age. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.905694.

Evidence status: Sourced tertiary total; cross-source allocation proxy . 18-24 demand proxy omits mature-age/international composition; university expenses include research and private funding; source concepts differ .

Cost rule: input_mix ; wage weight 55.473 %; Observed input share; scope proxy . Finance 2024 Table 4: employee benefits 23,891,495 / total expenses 43,068,392 ($000) .

Cost source and limitation: https://www.education.gov.au/download/19832/finance-2024-financial-reports-higher-education-providers/43005/2024-university-finance-summary-information/pdf ; Whole-provider structure includes research/international activity; not a pure government-funded teaching cost share .

e_vet: Vocational education and training

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25/55120DO095_202425.xlsx; Department of Education Finance 2024 Table 4; RoGS 2026 Table 5A.1 ; ABS tertiary control allocated using 2024 university total expenses versus VET recurrent expenses excluding user cost of capital .

Allocation and demand: 2025 $7.73bn; reconciled $7.73bn; raw 2030 $9.86bn; final 2030 $8.93bn. Demand profile: vet_age. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.905694.

Evidence status: Sourced tertiary total; cross-source allocation proxy . 15-64 demand proxy; allocation is not an exact GFS split; government-funded training participation and hours held constant .

Cost rule: funding_proxy ; wage weight 69.750 %; Published funding rule; scope proxy . National Skills Agreement flexible funding: 85% WCI-1 and 15% WCI-6; wage weights 75% and 40% .

Cost source and limitation: https://federalfinancialrelations.gov.au/sites/federalfinancialrelations.gov.au/files/2023-12/national-skills-agreement.pdf ; Commonwealth flexible-funding formula extended as proxy to all VET; productivity discount set to zero, not estimated .

e_other: Other education

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25/55120DO095_202425.xlsx ; Table_1 row 11 2024-25 .

Allocation and demand: 2025 $16.34bn; reconciled $16.34bn; raw 2030 $20.94bn; final 2030 $18.96bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.905694.

Evidence status: Sourced allocation; model demand proxy . Residual purpose includes preschool/other education; total-population proxy, not a fitted age profile .

Cost rule: input_mix ; wage weight 80.000 %; Retained judgment . Previous 80% education exposure retained for heterogeneous residual .

Cost source and limitation: Retained explicit scenario judgment ; Includes preschool and residual functions; not independently estimated .

a_residential: Residential aged care

Source and extraction: https://www.gen-agedcaredata.gov.au/topics/spending-on-aged-care ; Figure 1, government expenditure 2024-25; published rounded AUD bn .

Allocation and demand: 2025 $24.20bn; reconciled $24.20bn; raw 2030 $36.50bn; final 2030 $33.64bn. Demand profile: aged_care. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Uses common IGR aged-care age profile; distinct residential/home utilisation profiles not yet estimated .

a_home: Home care and support

Source and extraction: https://www.gen-agedcaredata.gov.au/topics/spending-on-aged-care ; Figure 1, government expenditure 2024-25; published rounded AUD bn .

Allocation and demand: 2025 $12.90bn; reconciled $12.90bn; raw 2030 $19.45bn; final 2030 $17.93bn. Demand profile: aged_care. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Uses common IGR aged-care age profile; distinct residential/home utilisation profiles not yet estimated .

a_flexible: Flexible aged care

Source and extraction: https://www.gen-agedcaredata.gov.au/topics/spending-on-aged-care ; Figure 1, government expenditure 2024-25; published rounded AUD bn .

Allocation and demand: 2025 $1.20bn; reconciled $1.20bn; raw 2030 $1.81bn; final 2030 $1.67bn. Demand profile: aged_care. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Uses common IGR aged-care age profile; distinct residential/home utilisation profiles not yet estimated .

a_other: Other aged care

Source and extraction: https://www.gen-agedcaredata.gov.au/topics/spending-on-aged-care ; Figure 1, government expenditure 2024-25; published rounded AUD bn .

Allocation and demand: 2025 $1.50bn; reconciled $1.50bn; raw 2030 $2.26bn; final 2030 $2.08bn. Demand profile: aged_care. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Uses common IGR aged-care age profile; distinct residential/home utilisation profiles not yet estimated .

p_age: Age Pension

Source and extraction: https://www.anao.gov.au/work/performance-audit/administration-of-the-age-pension ; 2024-25 outlay used in existing module .

Allocation and demand: 2025 $62.20bn; reconciled $62.20bn; raw 2030 $85.13bn; final 2030 $78.46bn. Demand profile: pension. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced outlay; synthetic entitlement model . Wealth/household archetypes, indexation approximations and cash-to-GFS mapping retained .

n_ndis: NDIS participant plans

Source and extraction: https://www.ndis.gov.au/media/8108/download ; Annual Report Note 1.1G; participant plan expenses .

Allocation and demand: 2025 $46.35bn; reconciled $46.35bn; raw 2030 $61.13bn; final 2030 $56.34bn. Demand profile: ndis. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced plan expense; existing cohort model . NDIS pass-through/intensity settings unchanged; administration remains in broader residual .

s_other: Other social protection (unallocated programmes)

Source and extraction: ABS GFS Social Protection control less separate pension/care/NDIS allocations ; Residual identity .

Allocation and demand: 2025 $163.39bn; reconciled $163.39bn; raw 2030 $206.52bn; final 2030 $190.34bn. Demand profile: working_age. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Residual allocation; working-age proxy . Not a separate simulation of JobSeeker, disability pensions, family assistance, child care or state welfare; remains a material modelling limitation .

o_1: General public services excluding conventional interest

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: General public services .

Allocation and demand: 2025 $64.03bn; reconciled $64.03bn; raw 2030 $81.55bn; final 2030 $75.17bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_2: Public order and safety

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Public order and safety .

Allocation and demand: 2025 $52.77bn; reconciled $52.77bn; raw 2030 $67.22bn; final 2030 $61.95bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_3: Economic affairs

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Economic affairs .

Allocation and demand: 2025 $46.37bn; reconciled $46.36bn; raw 2030 $59.05bn; final 2030 $54.43bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_4: Environmental Protection

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Environmental Protection .

Allocation and demand: 2025 $20.78bn; reconciled $20.78bn; raw 2030 $26.46bn; final 2030 $24.39bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_5: Housing and community amenities

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Housing and community amenities .

Allocation and demand: 2025 $13.90bn; reconciled $13.90bn; raw 2030 $17.71bn; final 2030 $16.32bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_6: Recreation, culture and religion

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Recreation, culture and religion .

Allocation and demand: 2025 $19.97bn; reconciled $19.97bn; raw 2030 $25.44bn; final 2030 $23.44bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

o_7: Transport

Source and extraction: https://www.abs.gov.au/statistics/economy/government/government-finance-statistics-annual/2024-25 ; All-government Table_4: Transport .

Allocation and demand: 2025 $47.16bn; reconciled $47.16bn; raw 2030 $60.06bn; final 2030 $55.36bn. Demand profile: population. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced allocation; model demand proxy . Total-population demand proxy; parent cost rule retained; functional expenses include depreciation/capital transfers; gross capital formation is not added .

d_defence: Defence

Source and extraction: ABS GFS Defence; prototype target ; Table_4; target reaches 2.5% GDP by 2035 .

Allocation and demand: 2025 $50.17bn; reconciled $50.17bn; raw 2030 $60.38bn; final 2030 $55.65bn. Demand profile: target. Parent reconciliation factor: 1.000000; 2030 anchor factor: 0.921653.

Evidence status: Sourced starting expense; policy judgment . GDP target overrides demand/cost equation; not a sourced defence projection .

12. Reproduction, checks and the next assumption discussion

Run scripts/acquire_bottom_up_sources.R once to cache additional official sources, then scripts/prepare_bottom_up_inputs.R offline. Run tests/validate_bottom_up.R and scripts/build_spending_background.R from the project or via Rscript from its parent directory. The report generator writes its HTML, PNG figures and CSV walk-through tables as well as this PDF. Annual subfunction and factor exports are available from Shiny.

The 64 bottom-up checks cover controls, unchanged anchors, NDIS/pension preservation, aggregation, factor identities and financing stress. A further 57 health/education checks verify observed-share arithmetic, separate rules, higher/VET demand, uniform scaling, zero-productivity scenarios and unchanged other categories. These establish implementation consistency, not forecast accuracy.

File	Purpose
data/spending_subfunctions.csv	Source allocations, profile mapping, locators and limitations
data/processed/subfunction_age_profiles.csv	Separate health age profiles extracted from IGR C12.10
R/spending_bottom_up.R	Projection, aggregation and exact contribution functions
outputs/spending_background/calibration.csv	All source-to-GFS-to-2030 reconciliation calculations
outputs/spending_background/annual_subfunctions.csv	Annual spending and multiplicative factors for all 24 blocks
outputs/spending_background/contributions_*.csv	Factor contributions by block, category and measure
archive/pre_bottom_up_2026_10_27	Frozen prior spending implementation and report

The next discussion should prioritise: age-specific enrolments and tertiary participation; separate residential/home care utilisation; splitting residual social protection and other health; measured wage/non-wage exposure; service-efficiency trends; and residual intensity with policy breaks removed. The present pass-through and intensity controls remain scenario judgments while that evidence is assembled. No new statistical uncertainty bands are claimed.

Primary source links

- PBO National Fiscal Outlook 2026-27 : consolidated anchor; workbook Table 1/2.
- ABS GFS 2024-25 : national Table 4 controls; Education workbook Table_1.
- 2026 IGR chart data : C8.6 population shares, C9.13 participation, C12.10 health profiles, C6.6 aged-care profile, C12.19 pension starting status.
- AIHW Commonwealth government health spending 2024-25 ; state own-source spending : rounded area allocations, Figures 11 and 16; PDF report and spreadsheets not required for the manually recorded figures.
- GEN aged-care expenditure : Figure 1, 2024-25, updated 28 August 2026.
- NDIA Annual Report 2024-25 ; Q4 participation report : Note 1.1G and Table D.13.
- ANAO pension outlay ; individual statutory parameters and dated URLs remain in data/policy_parameters.csv.

Additional source snapshot manifest

file	snapshot_date	bytes	md5
abs_education_2025.xlsx	2026-10-27	275,472	505923434492f23c32228e340c5d2789
aihw_commonwealth_2025.html	2026-10-27	219,528	0115e2a6093f0be3754063fb64a6d360
aihw_states_2025.html	2026-10-27	200,709	5713932e09e64b2a529cb14c574e5c98

file	snapshot_date	bytes	md5
gen_spending_2025.html	2026-10-27	216,518	9de362de75092f1352542a83135caf58
hospital_costs.xlsm	2026-10-27	1,261,735	c2b02f1c68d69cafb5588334b574fd41
hospital_report.pdf	2026-10-27	1,562,175	eb61bd3ae7dc4313c80b8332c6cce4f0
school_funding.pdf	2026-10-27	310,780	a82fdd41a86c95556c17966704b6c61b
vet_tables.xlsx	2026-10-27	537,783	6920793e3ffd051107d62491b928597c
university_finance.pdf	2026-10-27	760,849	6cf2c568bbd3d980e97fbbd6072f85ad
skills_agreement.pdf	2026-10-27	2,192,287	d95f202b93b06e7e66d92c512453c4ba

Prepared from the local e61 fiscal model, 27 October 2026. This background paper is a reproducible explanation of a research scenario model; it is not an official PBO, IGR, AIHW or NDIA forecast.